Neural-ANOVA: Analytical Model Decomposition using Automatic Integration

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Motivation

Functional ANOVA (fANOVA) decomposes any square-integrable function into orthogonal interaction terms, revealing how much each input, and every combination of inputs, contributes to the output.

Benefits

- interpretability: isolates main and higher-order interaction effects, enabling validation with domain experts
- universality: model-agnostic, applicable to any square integrable function.
- uncertainty analysis: fANOVA yields variance decomposition, useful for Design-of-Experiments (DoE) and risk analysis

Introduction: Functional ANOVA decomposition

fANOVA decomposes a function $f : [0,1]^K \to \mathbb{R}$ into the sum of low-dimensional terms

1. function decomposition

$$f(\boldsymbol{x}) = f_0 + \sum_{k=1}^{K} f_k(x_k) + \sum_{k=1}^{K} \sum_{l<k} f_{k,l}(x_k, x_l) + \cdots = \\ := \sum_{\mathcal{S}\subseteq\mathcal{K}} f_{\mathcal{S}}(\boldsymbol{x}_{\mathcal{S}}).$$

2. construction algorithm based on recursive integration over subspaces[a]

$$f_{\mathcal{S}}(\boldsymbol{x}_{\mathcal{S}}) = \int_{[0,1]^{K-|\mathcal{S}|}} f(\boldsymbol{x})\, d\boldsymbol{x}_{\mathcal{K}\setminus\mathcal{S}} - \sum_{\mathcal{U}\subsetneq\mathcal{S}} f_{\mathcal{U}}(\boldsymbol{x}_{\mathcal{U}}), \tag{1}$$

3. characteristics
   - decomposition is minimal, not introducing unnecessary terms
   - terms are orthogonal, $\langle f_1(x_1), f_{1,2}(x_1, x_2)\rangle = 0$
   - yields full decomposition into 2^K-terms

Calculating integrals with NNs

conventional NN training

$$\text{Loss}(\boldsymbol{\theta}) = \sum_i \phi\Big(f(x_1^{(i)}, x_2^{(i)}) - \text{NN}_{\boldsymbol{\theta}}(x_1^{(i)}, x_2^{(i)})\Big) \tag{2}$$

NN training to calculate integrals[b]

1. training the *autodiffed NN*

$$\text{Loss}(\boldsymbol{\theta}) = \sum_i \phi\Big(f(x_1^{(i)}, x_2^{(i)}) - \underbrace{\frac{\partial^2}{\partial x_1 \partial x_2}\text{NN}_{\boldsymbol{\theta}}(x_1^{(i)}, x_2^{(i)})}_{\text{calculated by autodiff}}\Big) \tag{3}$$

2. calculate integral by signed sum of multiple NN evaluations

$$\int_{l_1}^{u_1}\int_{l_2}^{u_2} f(x_1, x_2)\, dx_1 dx_2 = \text{NN}_{\boldsymbol{\theta}^\star}(l_1, l_2) - \text{NN}_{\boldsymbol{\theta}^\star}(u_1, l_2) - \text{NN}_{\boldsymbol{\theta}^\star}(l_1, u_2) + \text{NN}_{\boldsymbol{\theta}^\star}(u_1, u_2) \tag{4}$$

$$:= \text{NN}_{\boldsymbol{\theta}^\star}(x_1, x_2)\big|_{x_1,x_2\in(l_1,u_1)\times(l_2,u_2)} \tag{5}$$

3. generalization for integration over subsets of variables (e.g. 3-dim)

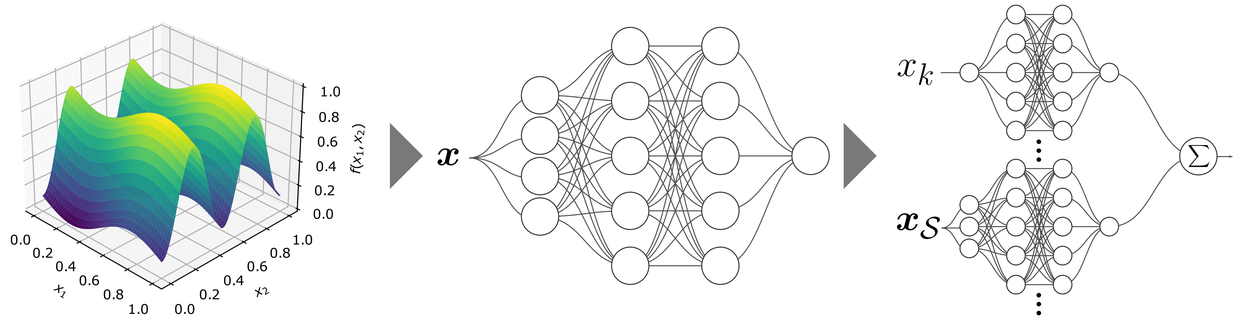
$$\int_{l_1}^{u_1} f(x_1, x_2, x_3) dx_1 = \frac{\partial^2}{\partial x_2 \partial x_3}\text{NN}(\boldsymbol{x})\Big|_{x_1\in(l_1,u_1)} \tag{6}$$

$$\int_{l_2}^{u_2}\int_{l_3}^{u_3} f(x_1, x_2, x_3) dx_2 dx_3 = \frac{\partial}{\partial x_1}\text{NN}(\boldsymbol{x})\Big|_{x_2,x_3\in(l_2,u_2)\times(l_3,u_3)}. \tag{7}$$

[a] *On decompositions of multivariate functions*, F. Kuo et al., Mathematics of computation, 2010

[b] *Autoint: Automatic integration for fast neural volume rendering, D. Lindell et al., CVPR 2021*

Neural-ANOVA decomposition



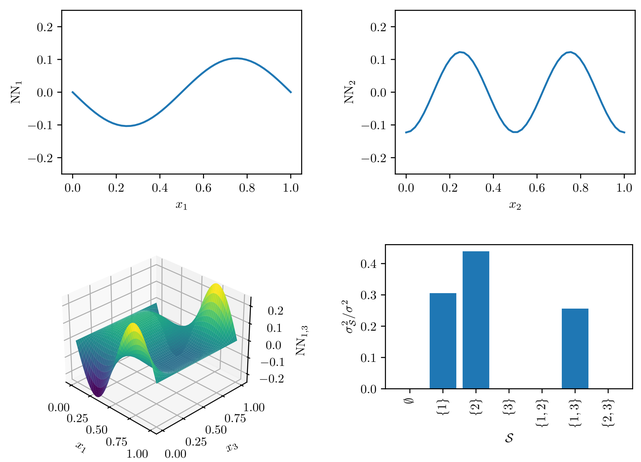
$$f(\boldsymbol{x}) \quad \approx \quad \frac{\partial^K}{\partial x_1 \cdots \partial x_K}\text{NN}(\boldsymbol{x}) \quad = \quad \sum_{\mathcal{S}} \text{NN}_{\mathcal{S}}(\boldsymbol{x}_{\mathcal{S}})$$

Algorithmic Steps

1. fit mixed-partial of NN to data $\text{NN}(\boldsymbol{x}) := \text{NN}_{\boldsymbol{\theta}^*}(\boldsymbol{x})$ by $f(\boldsymbol{x}) \approx \frac{\partial^K}{\partial x_1 \cdots \partial x_K}\text{NN}_{\boldsymbol{\theta}^*}(\boldsymbol{x})$
2. calculate integral over all variables $\text{NN}_{\emptyset} := \sum_{\boldsymbol{x}\in(0,1)^K}(-1)^s \text{NN}(\boldsymbol{x})$
3. calculate interactions recursively $\text{NN}_{\mathcal{S}}(\boldsymbol{x}_{\mathcal{S}}) := \frac{d^{|\mathcal{S}|}}{d\boldsymbol{x}_{\mathcal{S}}}\text{NN}(\boldsymbol{x})\big|_{\boldsymbol{x}_{\mathcal{S}^c}\in(0,1)^{K-|\mathcal{S}|}} - \sum_{\mathcal{U}\subsetneq\mathcal{S}} \text{NN}_{\mathcal{U}}(\boldsymbol{x}_{\mathcal{U}})$

Example: Ishigami test function

$$f(\boldsymbol{x}) = \sin(x_1) + a\sin^2(x_2) + bx_3^4\sin(x_1) \approx \frac{\partial^3}{\partial x_1 \partial x_2 \partial x_3}\text{NN}(x_1, x_2, x_3), \quad a = 7, b = 0.1 \tag{8}$$



Main Results: Neural-ANOVA is expressive and noise robust

- allows analysis of interaction effects and model bias for different architectures
- stable training for full-differentiation of two-layer MLPs up to 10 dimensions
- swish+sigmoid retain expressivity and noise robustness under mixed-differentiation
- relu collapses under differentiation, rectified polynomial (rep) favorable for sparse data
- code available at https://github.com/stli/neural-anova

